Scheme 4

Stage 3 Retrieval and Manual Relevance Triage Scheme

Full-text availability, retrieval need, and adjudication checklist

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DRAFT FOR EXPERT EVALUATION. These coding schemes are a working draft circulated for expert evaluation. They have not been applied to a final review corpus. The current manuscript is a test run that exercised an earlier, coarser generation of these schemes in the Python workflow with an LLM (gemini-2.5-flash); the full end-to-end run is deliberately held until after expert feedback, and the refinements proposed here are awaiting sign-off before any re-run.

At a Glance

Workflow position
Bridge between Stage 2 coding and Stage 3 full-text coding.

Operational mode
Automated candidate manifest plus human checklist completion.

Unit of analysis
One Stage 3 candidate record and its retrieval state.

Provenance basis
Manifest-generation logic in stage3_prep.py and the generated manual relevance checklist.

Research questions
Retrieval and adjudication gate that protects RQ2 and RQ3 corpus quality

Tagline
Automated candidate manifest plus human checklist completion

Canonical Source Paths

- `src/bps_review/extraction/stage3_prep.py`
- `src/review_stages/04_extraction/forms/stage3_manual_relevance_checklist.csv`
- `src/review_stages/04_extraction/outputs/stage3_candidate_manifest.csv`
- `src/review_stages/04_extraction/outputs/stage3_manual_fulltext_queue.csv`
- `src/review_stages/04_extraction/outputs/stage3_retrieval_validation.csv`

Purpose

This scheme governs the transition from Stage 2 abstract coding to Stage 3 full-text work. It standardizes which candidate reviews need manual retrieval, which records require manual relevance adjudication, and how retrieval status, risk signals, and reviewer decisions are recorded before deep coding begins.

It is not a logistics file. It is a standardized adjudication framework that decides whether a review can enter Stage 3, whether the retrieved text is adequate, and how problematic or ambiguous records are escalated for human judgment.

Manifest Fields

Automated fields written per candidate.

fulltext_status
Operational retrieval status of the candidate.

- **manual_retrieval_required** No machine-available full text was secured; human retrieval is needed.
- **pmc_open_available_not_cached** A PMCID exists but the file was not yet cached.
- **pmc_fulltext_cached** A PMC full text is already cached locally.
- **pmc_fulltext_fetched** A PMC full text was retrieved during the current run.
- **pmc_linked_fetch_failed** A PMCID link existed but retrieval or XML parsing failed.
- **pmc_fulltext_low_content_manual_check** Text was fetched but appears too short and needs manual inspection.

retrieval_source
Source of the PMCID or full-text link (existing metadata, PubMed elink, or Europe PMC).

fulltext_word_count
Word count of cached text, used to detect low-content full texts. *(free text)*

manual_retrieval_needed
Whether a human must retrieve the text.

- **yes** Retrieval required.
- **no** Machine text is available.

manual_relevance_priority
Adjudication urgency.

- **high** Withdrawal or missing pain focus.
- **medium** Other concern flags, or retrieval required with otherwise low signal.
- **low** No concern flags.

manual_relevance_flags
Pipe-delimited signal list describing why adjudication is needed. See the signal logic below. *(free text)*

osf_manual_adjudication_required
Currently set to yes for the checklist workflow.

cached_xml_path / cached_text_path
Relative cache paths for machine-fetched PMC files.

pubmed_url
Direct URL to the source record when a PMID is available.

Manual Relevance Signal Logic

Each signal is derived from title and abstract and drives priority.

`withdrawn_or_retracted_signal`
Assigned when the title or abstract indicates a withdrawn or retracted record. Forces high priority.

`pain_focus_not_explicit`
Assigned when pain is not explicit in title or abstract. Forces high priority.

`chronicity_not_explicit`
Assigned when chronic, persistent, or long-term framing is absent.

`review_design_unclear`
Assigned when review design is missing or too vague.

Reviewer Checklist Fields

Human adjudication fields completed against the manifest.

`reviewer_decision`
Human reviewer decision after checking the candidate. *(free text)*

`reviewer_notes`
Notes supporting the reviewer decision. *(free text)*

`adjudication_decision`
Final adjudicated status after disagreement resolution. *(free text)*

`adjudication_notes`
Adjudication rationale. *(free text)*

Proposed Refinement: Explicit Retrieval-Adequacy Gate

PROPOSED REFINEMENT *Awaiting expert evaluation. Not yet applied to the pipeline.*

Refinement proposed for expert evaluation. The low-content flag currently relies on an implicit word-count heuristic. We propose a documented adequacy gate with a stated threshold (for example a minimum body word count and the presence of a results or discussion section) and an explicit adequacy value: adequate, borderline, or inadequate. Borderline and inadequate texts route to manual retrieval even when a file exists.

We also propose a reproducible priority matrix that maps the set of triggered signals to a priority level deterministically, so two coders reading the same manifest row derive the same priority. This removes the current tie-break ambiguity when several medium signals co-occur.

Test-Run Corpus State

Counts from the current test run, for context only.

Stage 2 included records
111

Stage 3 candidate manifest
92

Manual full-text queue
63

Reliability subset (capped at 20)
18

Primary Outputs Using This Scheme

- `src/review_stages/04_extraction/outputs/stage3_candidate_manifest.csv`
- `src/review_stages/04_extraction/forms/stage3_manual_relevance_checklist.csv`
- `src/review_stages/04_extraction/outputs/stage3_manual_fulltext_queue.csv`
- `src/review_stages/04_extraction/outputs/stage3_retrieval_validation.csv`